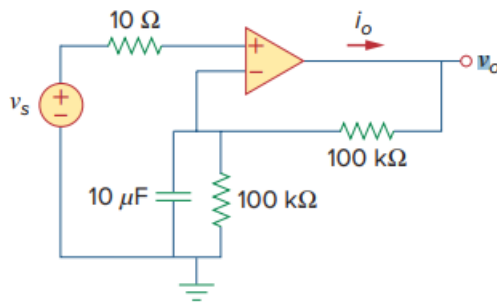


Problem

In the circuit of Fig. 7.140, find v_o and i_o , given that $v_s = 10[1 - e^{-t}]u(t)$ V.

Fig. 7.140:



Step-by-step solution

Step 1 of 3

Refer to Figure 7.140 in the textbook.

Redraw the circuit as shown in Figure 1.

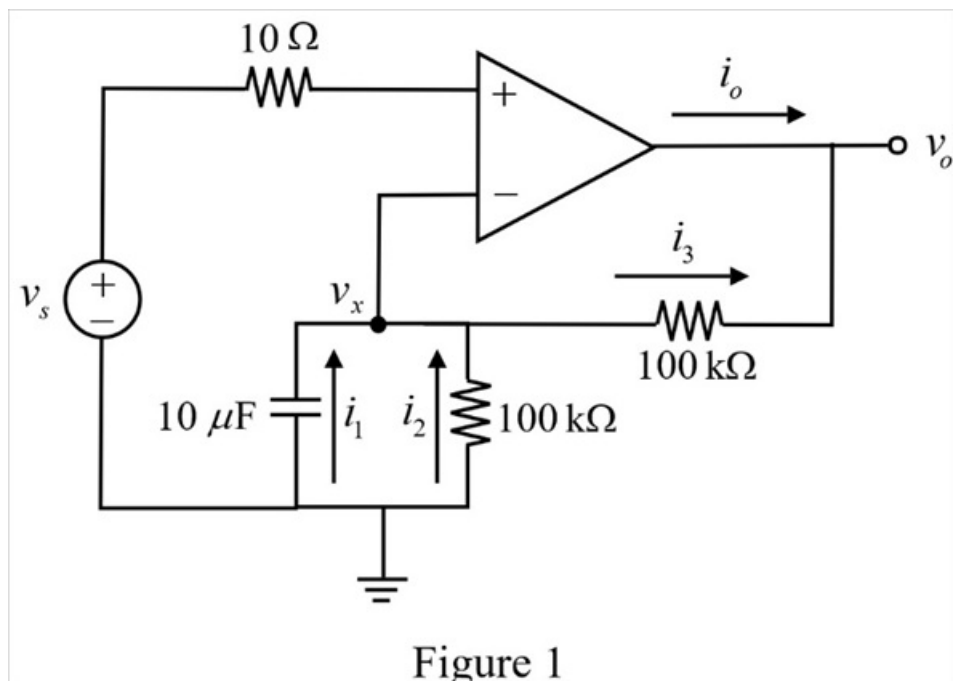


Figure 1

Step 2 of 3

Apply Kirchhoff's current law at node v_x .

$$i_1 + i_2 = i_3$$

$$C \frac{dv_x}{dt} + \frac{0 - v_x}{100 \times 10^3} = \frac{v_x - v_o}{100 \times 10^3}$$

$$C \frac{dv_x}{dt} + \frac{-2v_x}{100 \times 10^3} = \frac{-v_o}{100 \times 10^3}$$

$$v_o = 2v_x - (100 \times 10^3) C \frac{dv_x}{dt}$$

Since $v_x = v_s$ because of virtual short circuit, therefore,

$$v_o = 2v_s - (100 \times 10^3) C \frac{dv_s}{dt}$$

Substitute $10[1 - e^{-t}]u(t)$ for v_s and $10 \mu\text{F}$ for C .

$$\begin{aligned} v_o &= 2\{10[1 - e^{-t}]u(t)\} - (100 \times 10^3)(10 \times 10^{-6}) \frac{d}{dt}\{10[1 - e^{-t}]u(t)\} \\ &= [20 - 20e^{-t}]u(t) - 10(-e^{-t})u(t) \\ &= (20 - 10e^{-t})u(t) \text{ V} \end{aligned}$$

Hence, the output voltage (v_o) is $\boxed{(20 - 10e^{-t})u(t) \text{ V}}$.

Step 3 of 3

Determine the current i_o , which is equal to the current i_3 .

$$i_o = \frac{v_o - v_s}{100 \times 10^3}$$

Substitute $10[1 - e^{-t}]u(t)$ for v_s and $(20 - 10e^{-t})u(t)$ for v_o .

$$\begin{aligned} i_o &= \frac{(20 - 10e^{-t})u(t) - 10[1 - e^{-t}]u(t)}{100 \times 10^3} \\ &= \frac{10}{100 \times 10^3} \\ &= 100 \mu\text{A} \end{aligned}$$

Hence, the output current i_o is $\boxed{100 \mu\text{A}}$.